STAFF REVIEW · H446 309 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. To reach the base case.

2. The middle index of remaining interval.

3. No candidates remain; not found.

4. [2,8,6].

I Do — source worked steps

1. Explain the result — The n==0 base case stops recursion and returns zero.

We Do — guided CHECK

1. Explain the result
The boundaries exclude a failed midpoint, so the remaining interval shrinks.

You Do — source mark schemes

1. Unbounded calls [2], possible stack overflow [1]; correct base [2], parameter changes towards base [1].

2. low0 high4 mid2→5 [2], low3 high4 mid3→7 [1], low4 high4 mid4→9 [2], returns index4 [1].

3. Typical array merge needs O(n) peak auxiliary storage [3]; worst-case time O(n log n) [2]; temporary lifetime does not change peak Big O memory [1].

Exit Ticket — source answer

1. False. Test edge cases and varied inputs.